DHRUV PANWAR

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EDUCATION

Vellore Institute of Technology (VIT) 2023 – 2027
B.Tech, Computer Science Engineering (E-Commerce Technology) — **CGPA: 8.16/10**

SS Children's Academy, Moradabad, U.P. (CBSE) 2022 – 2023
Class XII (PCM) — **83.8%**

Podar International School, Ahmedabad (CBSE) 2021 – 2022
Class X — **88.4%**

TECHNICAL SKILLS

- **Languages:** C++ (Programming), Python (Project Development), SQL
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Design, Operating Systems, DBMS, Complexity Analysis
- **Competitive Programming:** 400+ problems solved across LeetCode, Codeforces, and CodeChef
- **Tools & Platforms:** Git, GitHub, Docker, FastAPI, CLI, Streamlit, Jupyter
- **AI/ML (Applied):** FastEmbed (ONNX), LangChain, FAISS, YOLOv8, OpenCV, scikit-learn, NumPy, Pandas, Gemini API

PROJECTS

VisionMind AI: Intelligent Image Analysis Service | github.com/Dhruv-Panwar042/VisionMind-AI
Live Demo: *image-recognition-model-9l2e.onrender.com* · ***Tools:*** *Python · FastAPI · YOLOv8 · Docker · Google Gemini · OpenCV · ReportLab*

- Built and deployed a containerized computer vision microservice using FastAPI and Docker, exposing 12 REST endpoints with interactive Swagger documentation and sub-300ms CPU inference across 80 object classes at 640×640 resolution.
- Reduced end-to-end cloud latency from ~35s to ~1.2s on a shared 0.5 vCPU host by implementing browser-side canvas downscaling (97% payload reduction, slashing raw ~4MB uploads to ~120KB) and bundling in-memory annotations into a single request.
- Integrated Google Gemini for visual scene understanding, implementing sequential fallback routing to eliminate service interruptions from API rate limits (HTTP 429); generated spatial density heatmaps and streamed multi-page ReportLab PDF inspection briefs in <600ms.

DocuVault AI: Grounded Multi-Document RAG Microservice | github.com/Dhruv-Panwar042/DocuVault-AI
Live Demo: *docuvault-ai-ah83.onrender.com* · ***Tools:*** *Python · FastAPI · LangChain · FAISS · FastEmbed (ONNX) · Docker · ReportLab*

- Engineered a decoupled RAG microservice with 1,000-character semantic chunking (200-char overlap) and in-memory FAISS indexing, parsing multi-PDF uploads directly from RAM byte buffers with zero temporary disk I/O latency.
- Optimized embeddings via local ONNX CPU inference (384-dimensional all-MiniLM-L6-v2 via FastEmbed), slashing peak indexing RAM by 53% (from ~500MB to ~235MB on 512MB hosts) to prevent container OOM crashes while delivering sub-150ms vector retrieval.
- Implemented sequential fallback routing across Gemini models, transparent L2 distance confidence tiers (High: ≤0.5, Medium: ≤1.0), an interactive chunk inspector, and automated ReportLab PDF briefs; validated via 7 automated tests (100% CI pass rate).

EXPERIENCE

Software Engineering Intern — MPOnline Limited (TCS Joint Venture) Remote | May 2026 – Jul 2026

- Applied agile software engineering practices across module refactoring and REST API workflows, maintaining version control across 15+ Git feature branches and validating core business logic with unit test suites achieving a 95%+ pass rate.

AI/ML Intern — MPOnline Limited (TCS Joint Venture) Remote | May 2026 – Jul 2026

- Engineered automated data preprocessing, anomaly detection, and feature extraction pipelines across structured tabular datasets of 10,000+ records, evaluating model performance across Precision, Recall, and F1-score metrics to establish high-accuracy predictive baselines.

CERTIFICATIONS & LANGUAGES

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — Oct 2025
- **NPTEL — Marketing Management II** — 96%, Top 2 Percentile (National Elite Certificate)
- **Languages:** English (Full Professional) · Hindi (Native) · Gujarati (Limited Working)